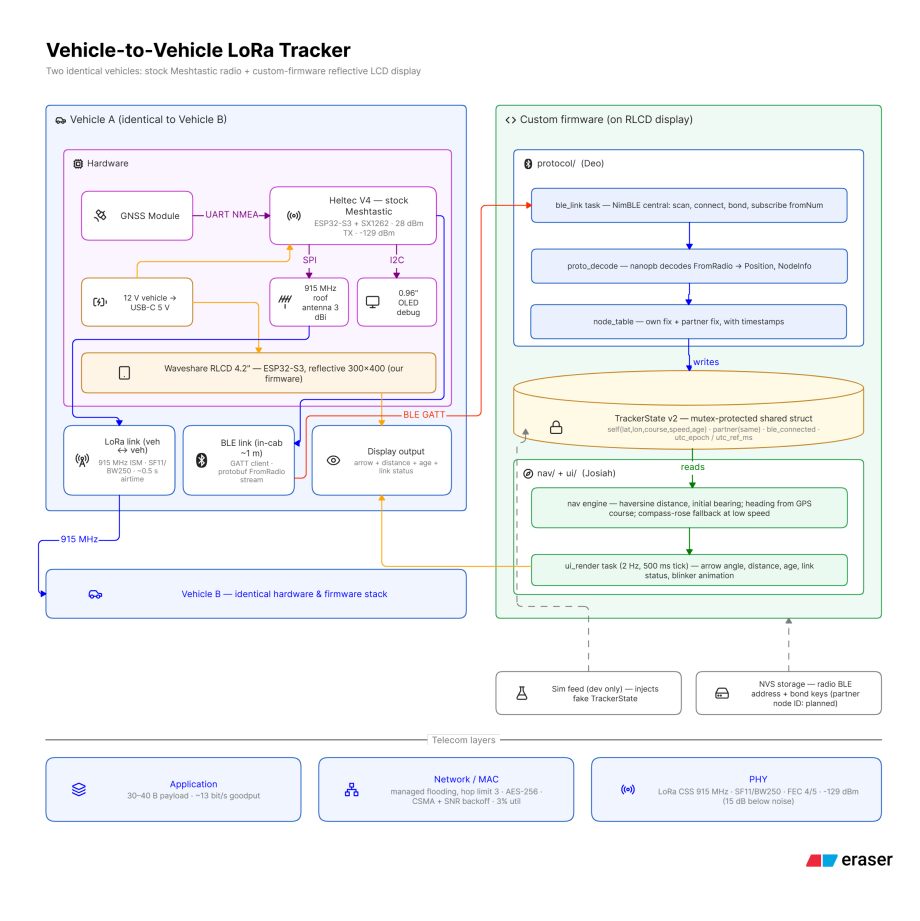
OFF-GRID VEHICLE TRACKER

TECHNICAL MANUAL, FOR ENGINEERS

Architecture, RF design with full derivations and worked numbers, firmware, navigation mathematics, provisioning, verification, and the complete catalog of failure modes learned on real hardware. Written so an engineer who has never seen this system can operate, debug, extend, and rebuild it.



GROUP 9 · CAPSTONE PROJECT LAB · TEXAS TECH UNIVERSITY · SPONSOR: DR. RICHARD GALE · REV 2 · 2026-09-30 · REPO: github.com/deosgracius/offgrid-vehicle-tracker

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1. System overview and the five ideas

Two Heltec WiFi LoRa 32 V4 radios (callsigns SF and RG) run unmodified Meshtastic firmware 2.7.26 and exchange position, telemetry, and text over a private 915 MHz LoRa channel. Each radio serves one Waveshare ESP32-S3-RLCD-4.2 driver display running our custom firmware, connected over Bluetooth Low Energy using the Meshtastic Client API [2]. The displays decode the radio traffic into a live navigation UI. There is no phone, tower, or internet dependency anywhere in the chain.

Five ideas explain essentially every behavior and every failure this system has ever shown. They recur throughout this manual.

IDEA 1. AGREEMENT IS THE CONNECTION. LoRa has no pairing or session. Two radios exchange data exactly when they independently agree on frequency, modulation, and encryption key. Every "total silence" failure in this project was one of those three quietly disagreeing.

IDEA 2. THE NAME IS THE FREQUENCY. Meshtastic derives the operating frequency from a hash of the primary channel name [2]. Rename the channel and the radio retunes. The OLED LoRa screen shows the synthesized frequency and is the ground truth; configuration read-backs can lie (Section 12, entries 1 and 6).

IDEA 3. DECIBELS ARE RATIOS, AND LORA DECODES BELOW THE NOISE. Signal quantities are logarithmic: +3 dB is double, dBm anchors to 1 mW. At spreading factor 11 the receiver decodes down to 17.5 dB below the noise floor [1], because each bit is spread over 2048 chips that integrate coherently.

IDEA 4. ONLINE IS ARITHMETIC, NOT A CONNECTION. Every liveness indicator is a comparison between now and a stored last-heard timestamp [3]. Every counter lie we chased was a timestamp problem, never a radio problem.

IDEA 5. AIRTIME IS THE BUDGET. Each packet costs roughly half a second of channel time (Section 3.3); the firmware throttles position broadcasts above 25 percent utilization [2]. Design every periodic packet to name the failure it prevents.

2. Hardware inventory and identification

Unit	Identity	Key facts
Radio SF	Node `!49b5ead4`, BLE MAC `8c:fd:49:b5:ea:d5`	Heltec V4: ESP32-S3 MCU, Semtech SX1262 transceiver [1], u-blox GNSS, OLED status screen, lithium cell. Pairs with Display 1. `bluetooth.mode NO_PIN`.
Radio RG	Node `!49b5bb5c`, BLE MAC `8c:fd:49:b5:bb:5d`	Same hardware. Pairs with Display 2. `bluetooth.mode NO_PIN`.
Display 1	USB serial `A4:CB:8F:D0:7D:40`	Waveshare ESP32-S3-RLCD-4.2: reflective LCD 400x300, sunlight readable, no backlight; runs our firmware.
Display 2	USB serial `94:A9:90:DD:14:3C`	Identical firmware; remembers RG in NVS after its first scan.

Flashing trap: radios and displays share USB VID:PID 303A:1001. Radio MACs begin `8C:FD:49`. Always check the `SER=` field of the COM port before any upload. Flashing a radio with display firmware erases its Meshtastic and requires factory recovery (Section 12, entry 8).

Enclosures: radios ride in community-printed cases; the display case is the AWSW "Symmetrical case" (Printables model 1610004, CC BY-NC 4.0) [6], printed in PETG for dashboard heat tolerance, fastened with four M2.5x6 screws. Six millimeters maximum: longer screws press on the panel glass.

3. RF design: the contract, the slot math, the modulation

3.1 The contract (Idea 1)

Layer	Our value	If mismatched
Frequency	Slot 43 = 912.875 MHz, derived from channel name "Group9"	Radios tuned apart; zero coupling
Modulation	LONG_FAST preset: SF11, 250 kHz bandwidth, coding rate 4:5	Chirps undecodable
Channel encryption	AES-256 pre-shared key, distributed in the channel URL	Packets heard, payloads dropped
Node identity	Curve25519 public keys exchanged via NodeInfo	Direct messages fail silently or NAK

3.2 Frequency slot derivation (Idea 2)

$$f_{slot} = f_{start} + \frac{BW}{2} + [\text{djb2}(name) \bmod N_{slots}] \cdot BW, \qquad N_{slots} = \frac{f_{stop} - f_{start}}{BW} = \frac{26\ \text{MHz}}{250\ \text{kHz}} = 104$$

The hash is djb2: `h = 5381; for each character c: h = h*33 + c; slot = h mod 104`. The formula calibrates against a known constant: djb2("LongFast") mod 104 = 19, which is 906.875 MHz, the published US default [2]. For our channel:

$$f_{43} = 902.000 + 0.125 + 43 \times 0.250 = 912.875\ \text{MHz}$$

Consequence, proven the hard way: two radios with identical keys but different primary channel names sit megahertz apart and never couple. The public regional mesh lives on slot 19, so slot 43 carries only our traffic. The frequency is public airspace; privacy comes from the key (Section 5).

3.3 Modulation and time on air (Ideas 3 and 5)

LoRa is chirp spread spectrum: each symbol is a frequency sweep across the full channel, and the information is the sweep's starting offset. SF11 gives 2^{11} = 2048 offsets, hence 11 bits per symbol and a symbol time of:

$$T_{sym} = \frac{2^{SF}}{BW} = \frac{2^{11}}{250{,}000\ \text{Hz}} = 8.19\ \text{ms}$$

De-chirping integrates 2048 chips coherently, which is why the SX1262 demodulates at signal-to-noise ratios down to -17.5 dB at SF11 [1]. The coding rate 4:5 adds one parity bit per four data bits for burst-error repair. The payload symbol count follows Semtech's formula [4], with PL the payload bytes, H = 0 for an explicit header, DE the low-data-rate-optimize flag, and CR = 1 for rate 4:5:

$$N_{payload} = 8 + \max\left(\left\lceil \frac{8PL - 4SF + 28 + 16\,CRC - 20H}{4(SF - 2DE)} \right\rceil (CR + 4),\ 0\right)$$

Worked example, a 40-byte position packet (SF = 11, BW = 250 kHz, DE = 0, CRC on, 16-symbol preamble):

$$N_{payload} = 8 + \left\lceil \frac{8(40) - 44 + 28 + 16}{44} \right\rceil \times 5 = 48, \quad T_{pkt} = (16 + 4.25 + 48)\,T_{sym} = 68.25 \times 8.19\ \text{ms} \approx 0.56\ \text{s}$$

Roughly 0.56 seconds of air per position update. This single number drives the entire cadence and airtime design in Sections 4.2 and 9.

4. Link budget and airtime budget

4.1 Link budget

$$P_{rx} = P_{tx} + G_{tx} + G_{rx} - FSPL, \qquad FSPL = 20\log_{10} d + 20\log_{10} f - 147.55 \;\; (d \text{ in m},\ f \text{ in Hz})$$

Term	Value	Source
Transmit power P_tx	+30 dBm (1 W)	Config; US legal maximum for this band [5]
Antenna gains G_tx, G_rx	~0 dBi each	Stock quarter-wave whips
Receiver sensitivity S	-131 dBm	SX1262 datasheet at SF11 / 250 kHz [1]
Total budget P_tx - S	161 dB	Loss the link can absorb before decode fails

Worked example, free space at 913 MHz:

$$FSPL_{1\,\text{km}} = 60 + 179.2 - 147.55 = 91.7 \text{ dB} \Rightarrow P_{rx} = -61.7 \text{ dBm}; \qquad FSPL_{5\,\text{km}} = 105.7 \text{ dB} \Rightarrow P_{rx} = -75.7 \text{ dBm}$$

Even at 5 km the received signal (-75.7 dBm) sits 55 dB above sensitivity in free space. Streets are not free space: we reserve 20 dB for terrain, clutter, and fading, which lands the honest design range at 3 to 5 km. Measured baseline: round-trip traceroute at desk range reports 5.75 to 7.25 dB SNR per direction, and margin converts to distance by the doubling rule:

$$\frac{d_2}{d_1} = 10^{\Delta/20} \Rightarrow \Delta = 6 \text{ dB} \approx \times 2 \text{ distance}, \qquad \text{margin} = SNR_{meas} - SNR_{floor} = 6 - (-17.5) = 23.5 \text{ dB}$$

4.2 Airtime budget (Idea 5)

Meshtastic suppresses position broadcasts above roughly 25 percent channel utilization [2]. At vehicle cadence the arithmetic is:

$$U = \frac{N_{nodes} \cdot T_{pkt}}{T_{cadence}} = \frac{2 \times 0.56}{10} \approx 11\% \text{ (no relay)}; \quad \times 2 \text{ relayed} \approx 22\% \text{ vs the 25\% throttle}$$

The default mesh role relays every overheard packet (hop limit 3); with exactly two nodes every relay is waste and doubles the spend to roughly 22 percent, brushing the throttle precisely when both vehicles move. Both radios therefore run role `CLIENT_MUTE` (receive and send everything, relay nothing), measured to halve idle utilization from 9-13 percent to 5-6 percent. Revert both to `CLIENT` if a third node or repeater ever joins.

5. Security model

- **Channel traffic** (positions, broadcasts, canned texts): AES-256 with a 256-bit pre-shared key generated at channel creation and distributed only inside the provisioning URL over USB [2]. The key never crosses the air.
- **Direct messages and position requests**: additionally encrypted per node with Curve25519 public keys exchanged via NodeInfo. A stale stored key makes direct traffic fail one-way and silently (Section 12, entry 4).
- **What an eavesdropper on slot 43 learns**: packet headers are cleartext, so node IDs, timing, and signal levels are visible. Payloads are not. Existence and schedule leak; content and location do not.
- **Key custody**: the channel URL embedding the key lives in the private team repository by deliberate decision. Never publish it. Rotating to a fresh channel takes minutes and orphans the old key.

6. Radio configuration reference and provisioning

Setting	Value	Rationale
`channel (primary)`	Group9, position precision 32	Private slot 43; full-resolution positions (the channel default of 13 truncates every broadcast to a ~6 km grid)
`lora.region`	US	902 to 928 MHz plan [5]
`lora modem preset`	LONG_FAST	SF11 / 250 kHz / 4:5, the range versus rate compromise
`lora.tx_power`	30	Legal maximum; margin is range
`position.position_broadcast_secs`	60	Parked cadence with a fix
`position.broadcast_smart_minimum_interval_secs / _distance`	10 / 20	Moving cadence: 10 s AND 20 m must both pass
`position.gps_update_interval`	10	Must be ≤ the smart interval or stale fixes re-broadcast
`telemetry.device_update_interval`	300	Liveness floor; keeps counters honest at ~0.17 percent airtime
`device.role`	CLIENT_MUTE	Two-node mesh: no relaying (Section 4.2)
`bluetooth.mode`	NO_PIN	Just-Works pairing for the display; bonded-mode radios stop advertising and become unreachable
Canned messages	21 full-word presets	Hard firmware cap: the whole set is one string of at most 200 characters (ours uses 194)

Provisioning procedure

```
# The channel URL (containing the key) is in hardware/README.md. Never publish it.
python -m meshtastic --port COMx --seturl "<CHANNEL-URL>"
python -m meshtastic --port COMx --set position.position_broadcast_secs 60 \
  --set position.broadcast_smart_minimum_interval_secs 10 \
  --set position.broadcast_smart_minimum_distance 20 \
  --set position.gps_update_interval 10 \
  --set telemetry.device_update_interval 300 \
  --set device.role CLIENT_MUTE
python -m meshtastic --port COMx --set bluetooth.mode NO_PIN     # radio that serves a display
python -m meshtastic --port COMx --set-owner-short "SF"          # or "RG"
python -m meshtastic --port COMx --set-canned-message "OK|Yes|No|Hello|..."   # full set in hardware/README
```

Verification is part of the procedure, not optional: read every value back with `--get`, confirm the OLED LoRa screen reads 912.875, then power-cycle and confirm 912.875 survives. An unsaved channel write is the classic silent revert (Section 12, entries 1 and 6).

7. Display firmware architecture

PlatformIO project `firmware-display/`, pioarduino platform (Arduino core 3.x), roughly 2200 hand-written lines of C and C++ plus nanopb-generated protobuf code. One-direction dataflow, one writer and one reader around a

mutex-protected store:

```
BLE task (NimBLE central)                       UI task (2 Hz render)
 ble_link: scan/connect/pair/subscribe/drain     tracker_store_get()
   -> proto_decode (nanopb FromRadio)              -> dead_reckon projection (Sec. 8)
   -> node_table (positions, names, batteries,     -> nav_compute (haversine, bearing)
      texts, timestamps)                           -> ui_render (3 pages on the RLCD)
   -> tracker_store_set(TrackerState v5)
```

7.1 The contract: TrackerState (version 5)

`include/tracker_state.h` is co-owned by the protocol and UI sides and versioned; changes ship as one agreed commit. Pure data, no RTOS or Arduino dependencies, so it also compiles on the host for unit tests.

```
typedef struct { double lat_deg, lon_deg; float course_deg, speed_mps;
                 uint32_t age_s; bool valid; } Fix;
typedef struct {
  Fix self, partner;            // own vehicle; partner vehicle
  bool ble_connected;           // in-cab link to our radio
  uint32_t utc_epoch, utc_ref_ms;          // GPS wall clock + millis reference
  char self_name[8], partner_name[8];      // radio callsigns from NodeInfo
  uint32_t msg_seq, msg_ms;                // latest received text (popup)
  char msg_from[8], msg_text[40];
  uint8_t msg_hist_n;                      // received-message history, newest first
  TrackerMsg msg_hist[4];
  uint8_t self_batt, partner_batt;         // 0..100; 101 external power; 255 unknown
} TrackerState;
```

7.2 BLE link (Meshtastic Client API [2])

GATT item	UUID	Role
Service	`6ba1b218-15a8-461f-9fa8-5dcae273eafd`	Meshtastic Client API
ToRadio	`f75c76d2-129e-4dad-a1dd-7866124401e7`	Write: commands and outbound packets
FromRadio	`2c55e69e-4993-11ed-b878-0242ac120002`	Read until empty: config dump and inbound packets (firmware 2.x UUID)
FromNum	`ed9da18c-a800-4f66-a670-aa7547e34453`	Notify: "new data queued", triggers a drain

On connect: write `want_config_id`, drain the configuration dump (node DB, MyInfo), then drain on every notify. The radio advertises only from boot until its first BLE disconnect, so the product flow is ignition-on: power the radio and display together; the display fires direct connect attempts and catches the boot window. The radio's address persists in NVS; a scan that finds any Meshtastic service re-learns it, which is how replacement units pair automatically.

7.3 Decode and node table

Ports handled: POSITION_APP (wire speed is km/h; the display stores m/s; a 4.5 km/h deadband suppresses parked GPS jitter), NODEINFO_APP (names and public keys), TELEMETRY_APP (battery), TEXT_MESSAGE_APP (raw UTF-8). Two honesty mechanisms live here: a bit-identical position re-broadcast does not reset the data age (a fixless radio re-sending its stored fix must not look fresh), and texts from our own node number are discarded because the radio echoes sent messages over BLE and only the receiving display should notify.

7.4 UI features

Three pages (compass, status, messages; tap cycles; hold resets tracking, or clears history on the messages page). Callsigns everywhere. Link tag at 90 s freshness. Receiver-side message popup for 8 s plus a 4-deep history. Closing/apart/holding trend (Section 8). Partner battery banner at 20 percent. Hollow dart and inverted age past 30 s. The display's BOOT button transmits a POSITION_APP want-response packet to the partner: an on-demand position request identical to the phone app's.

8. Navigation mathematics

All navigation outputs are computed on the display from the two stored fixes. Angles use latitude φ and longitude λ in radians; ψ is own course over ground.

8.1 Distance: haversine

$$a = \sin^2\frac{\Delta\varphi}{2} + \cos\varphi_1\cos\varphi_2\sin^2\frac{\Delta\lambda}{2}, \qquad d = 2R\arcsin\sqrt{a}, \qquad R = 6371\ \text{km}$$

Worked example from the bench: RG at (33.58656, -101.87542), SF at (33.58646, -101.87536). Δφ = 0.00010°, Δλ = 0.00006°. The haversine gives d ≈ 12.5 m ≈ 41 ft, exactly what the status page showed. At this scale the formula degenerates gracefully to flat-earth Pythagoras; it is used because it stays accurate at every scale up to antipodes.

8.2 Bearing and the arrow

$$\theta = \operatorname{atan2}(\sin\Delta\lambda\cos\varphi_2,\ \cos\varphi_1\sin\varphi_2 - \sin\varphi_1\cos\varphi_2\cos\Delta\lambda), \qquad \alpha_{arrow} = \operatorname{wrap}_{\pm180}(\theta - \psi_{course})$$

θ is the initial great-circle bearing to the partner, drawn as the dart on the compass ring. While driving, the relative angle α (bearing minus own course, wrapped to ±180°) drives the blinker chevrons, the "at N o'clock" wording (each hour is 30 degrees), and the off-course warning. Course over ground comes from the GPS and is meaningful only at speed, which is why a stopped vehicle switches to compass wording: there is no nose to be relative to.

8.3 Dead reckoning between fixes

$$\varphi(t) = \varphi_0 + \frac{v\,\Delta t\cos\psi}{R}, \qquad \lambda(t) = \lambda_0 + \frac{v\,\Delta t\sin\psi}{R\cos\varphi}, \qquad \Delta t \le 120\ \text{s}$$

Between packets, both fixes are projected forward along their last course and speed, capped at 120 s of projection. The age counter is deliberately NOT reset by projection: the screen motion stays smooth while the age and the hollow dart tell the truth about data freshness.

8.4 Closing-speed trend

$$r_k = 0.9\,r_{k-1} + 0.1\cdot\frac{d_{k-1} - d_k}{\Delta t}, \qquad \Delta t = 0.5\ \text{s};\quad r > 0:\ \text{closing},\quad r < 0:\ \text{apart}$$

An exponential moving average of the distance derivative, sampled each 500 ms render tick. The 0.9/0.1 weighting settles in roughly five seconds, enough smoothing that GPS jitter does not flap the label. Displayed as "closing N mph", "apart N mph", or "holding" inside a 2 mph deadband; suppressed entirely while partner data is stale.

9. Timing, liveness, and honesty semantics (Idea 4)

$$\text{online}(n) \Leftrightarrow t_{now} - t_{last\ heard}(n) < 7200\ \text{s}$$

The 7200 s constant is stock firmware: `#define NUM_ONLINE_SECS (60 * 60 * 2)` in `src/mesh/NodeDB.cpp` of meshtastic/firmware [3], sized for an ecosystem where solar and battery nodes may legitimately stay silent for hours. The window must exceed the longest honest silence or healthy nodes flap offline; the cost is that a dead radio looks alive for up to two hours. Our traffic contract guarantees a packet at least every 300 s, so the display layers stricter thresholds:

Indicator	Threshold	Meaning
SF<->RG OK tag	heard ≤ 90 s	Link actively fresh
Hollow dart, inverted age	age > 30 s	Position projected by dead reckoning, not measured
DATA STALE banner	age > 120 s	Stop trusting the position
Radio OLED "online"	heard ≤ 7200 s	Stock ecosystem bound

Cadence design rule: every periodic packet names the failure it prevents. Position at 10 s / 20 m moving (tracking staleness), position at 60 s parked (presence), telemetry at 300 s (counter lies), NodeInfo on boot and periodically (names and keys).

10. Build, flash, and test procedures

```
# build + flash (ALWAYS pass --upload-port; check the SER= MAC first, Section 2)
pio run -e display-ble -t upload --upload-port COMx   # real BLE build
pio run -e display     -t upload --upload-port COMx   # SIM build: fake data, BOOT cycles scenarios
pio device monitor --port COMx --baud 115200
pio test -e native                                    # 12 nav unit tests on the host
```

- Board flags matter: pioarduino platform, `memory_type dio_opi`, flash mode `dio`. QIO silently boot-loops this panel. Do not "fix" it back.
- On the project workstation the native tests need MinGW on PATH: `PATH=/c/mingw64/bin:$PATH pio test -e native`.
- Every serial connection to a radio or display hard-resets it via DTR. Consequences: config writes can silently revert if the port closes mid-save (verify by read-back AND OLED), RAM-only node-table state is lost on each connect, and orphaned monitor processes hold COM ports (kill `device monitor` processes before flashing).
- Bring-up of a new display follows the checklist in `firmware-display/README.md`: SIM proof, BLE link, partner tracking, the position-request test (partner parked, age ≥ 30 s, press BOOT, expect the age to snap to seconds), and cased-versus-bare BLE RSSI within 2 dB.

11. Verification record

Claim	Evidence
Two-way link on Group9	Round-trip traceroutes across multiple sessions: 5.75 to 7.25 dB SNR per direction
Position agreement at rest	Two independent fixes 13 to 45 ft apart on one table (consumer GNSS scatter)
Live tracking	Display tracked a real drive on first boot; the 10 s moving cadence was proven in the Week 4 drive test
On-demand position request	CLI and display-button requests both returned the partner's live fix at full precision
Messaging	Canned texts delivered both directions; popup confirmed receiver-side only on hardware
Airtime claim	CLIENT_MUTE halved measured idle utilization from 9-13 to 5-6 percent
Unit tests	12/12 navigation-math tests pass on the native host build

Next planned verification: the two-vehicle drive test with metrics declared in advance (95th percentile partner-position age ≤ 15 s while moving; reacquisition after separation ≤ 60 s; zero silent drops in convoy), followed by SNR-versus-distance range characterization plotted against the Section 4 prediction.

12. Failure modes and recovery (the scar catalog)

Every entry below happened on this hardware and was reproduced or root-caused before being written down.

#	Symptom	Root cause	Recovery
1	Total mutual silence; configs read back identical; 0 percent channel utilization	Frequency slot mismatch: a channel write silently reverted, leaving the radios on different slots with the same key (Ideas 1 and 2)	Check both OLED LoRa screens: Group9 must read 912.875. Re-apply the channel URL, verify, power-cycle, verify again
2	Battery ran flat; the radio later boots looking healthy but config is factory or channels are lost	Battery-death crash reverts NVS configuration (observed: one radio lost channels only, one lost everything including keys and name)	Full provisioning recipe. Pre-flight rule: never run radios to empty; check 912.875 before every session
3	Screen lit, GPS fix shown, but the peer hears nothing for hours and USB serial refuses to connect	Half-alive wedge after a flat battery: the UI runs while LoRa TX and serial are frozen	Full power-cycle. Judge a radio by the OTHER radio's last-heard, never by its own screen
4	Broadcasts flow but direct messages and position requests time out in one direction, with no NAK	Stale public key: the peer's key regenerated in a factory event; the requester encrypts to the old key; the receiver drops silently	`--reset-nodedb` on the requester, wait one minute for NodeInfo relearn, retry
5	"Online" counters lie in either direction; entries stamped months old; distance shows kilometers between adjacent radios	Stale stored timestamps and positions: fixless radios re-broadcasting old fixes, flash-reloaded node records, clock-less boots (Idea 4)	Purge stored positions (`--remove-position`), sync clocks, reset node DBs. The 300 s heartbeat prevents recurrence
6	A config write appears to succeed; later reads show the old value	The serial port-close DTR reset rebooted the chip mid-save	Always verify by read-back AND the OLED; for stubborn values use the Python API holding the port open ~20 s
7	Node table shows nameless ghost entries; freshly learned entries vanish after a USB session	Crash-corrupted records; RAM state wiped by the connect reset before the scheduled flash save	`--reset-nodedb`, then leave the radio off USB; entries repopulate from live traffic
8	A Meshtastic radio erased by a firmware upload	An auto-picked COM port hit the radio (shared VID:PID)	esptool erase, factory `firmware-heltec-v4` binary at 0x0, then full provisioning; the node number survives (MAC-derived)

Appendix A: acronyms and abbreviations

Term	Expansion and meaning in this system
AES-256	Advanced Encryption Standard, 256-bit key; encrypts all channel traffic
BLE	Bluetooth Low Energy; the in-cab display-to-radio link at 2.4 GHz
BW	Bandwidth; RF channel width, 250 kHz here
CFR	Code of Federal Regulations; 47 CFR 15.247 governs this band
ChUtil	Channel utilization; percent of the last minute the channel carried RF energy
CR	Coding rate; forward-error-correction ratio, 4:5 here
CRC	Cyclic redundancy check; per-packet integrity checksum
dB / dBm / dBi	Decibel ratio / power referenced to 1 mW / antenna gain referenced to an isotropic radiator
DM	Direct message; addressed to one node, per-node encrypted
DTR	Data Terminal Ready; serial control line whose toggle hard-resets the ESP32
EMA	Exponential moving average; smooths the closing-speed trend
FCC	Federal Communications Commission; US radio regulator
FSPL	Free-space path loss; the distance term of the link budget
GATT	Generic Attribute Profile; BLE's service and characteristic structure
GNSS / GPS	Global navigation satellite system / the US constellation specifically
ISM	Industrial, Scientific, Medical; the license-free 902 to 928 MHz band
LoRa	Long Range; chirp spread spectrum modulation
MCU	Microcontroller unit; the ESP32-S3 in every device here
NAK / ACK	Negative acknowledgment / acknowledgment; delivery failed / delivered
NVS	Non-volatile storage; the ESP32 flash key-value store holding configuration
OLED	Organic LED; the radio's small status screen
PHY	Physical layer; the radio chip and modulation level of the stack
PKC	Public-key cryptography; Curve25519 pairs used for direct messages
PSK	Pre-shared key; the channel secret both radios hold
RLCD	Reflective liquid crystal display; sunlight-readable, backlight-free panel
RSSI	Received signal strength indicator; raw received power
SF	Spreading factor (context: modulation); also the callsign of radio 1
SNR	Signal-to-noise ratio; signal height above the noise floor, in dB
SMA	SubMiniature version A; the screw-on antenna connector
ToA	Time on air; the channel time one packet occupies
UUID	Universally unique identifier; names GATT services and characteristics
UTC	Coordinated Universal Time; the GPS-supplied wall clock

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